

Chapter 6

Haloalkanes & Haloarenes

⇒ Haloalkanes — Introduction, Nomenclature, Nature of Bond.

General formula: $C_nH_{2n+1}X$



Naming: name the alkyl group then halide

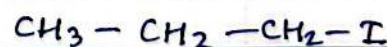
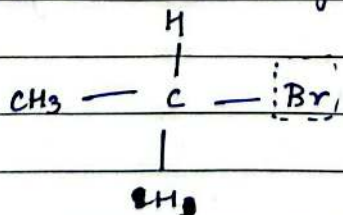
eg: CH_3Cl = methyl chloride

CH_3CH_2Br = ethyl bromide

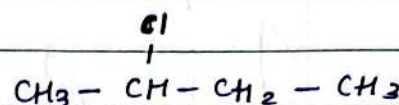
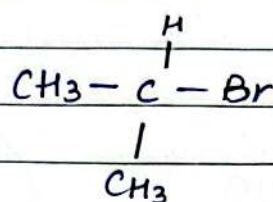
→ Classification of Alkyl Halides

a) Primary (1°) alkyl halides:

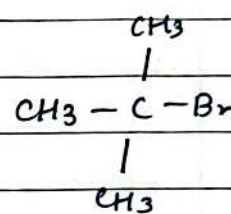
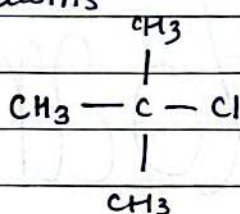
The haloalkanes in which the halogen atom is attached to a carbon which is not attached to any other carbon. or only one carbon.



b) Secondary (2°) alkyl halides: X ^{Halogen} on a carbon connected to two other carbon atoms



c) Tertiary (3°) Alkyl Halides: X ^{Halogen} on a carbon attached to three other carbon atoms



→ Nature of C-X bond.

The C-X bond is polar (X is more electronegative than C) i.e. with carbon carrying a partial positive charge. This makes carbon susceptible to nucleophilic attack.

Down the halogen group, atomic size increases. Bond length increases, bond energy decreases.

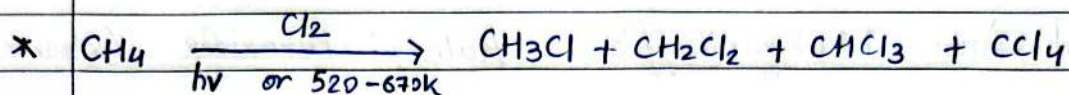
C-F → shortest / Strongest

C-I → longest / weakest

This is the reason why Alkyl iodides are most reactive and alkyl fluorides are least.

⇒ Preparation of Haloalkanes

i) From Alkanes : (Reacts with Cl and Br)



chloro
methane

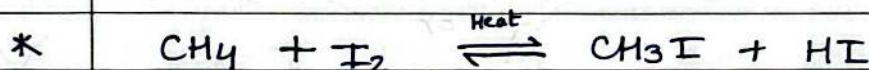
chloro
methane

chloro
methane

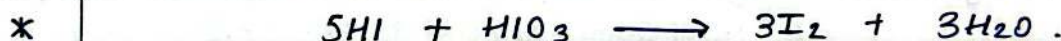
Trichloro
methane

Tetrachloro
methane

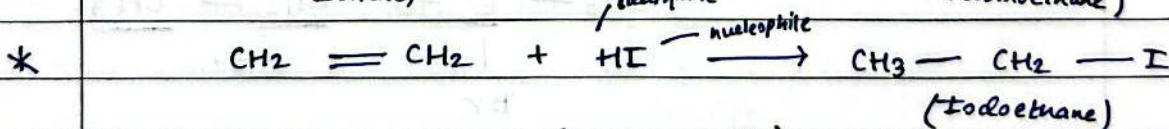
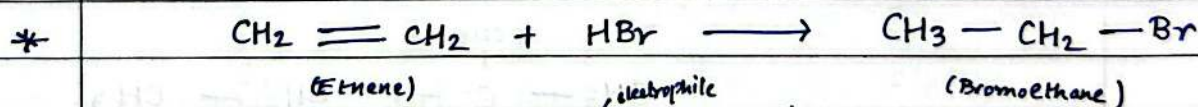
→ Not preferred in lab, since it does not stop at monosubstitution
Used at Industrial Level



→ Since HI is a strong reducing agent
It reduces CH_3I product straight back to CH_4 , the reaction does not go to completion on its own. To stop this a strong oxidizing agent such as HIO_3 or HNO_3 which oxidizes HI.



ii) From Alkenes — Addition of hydrogen halide

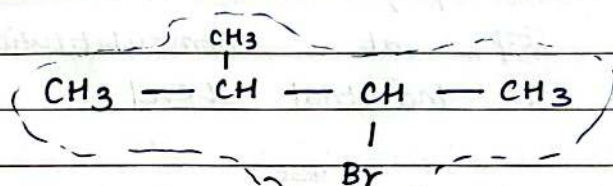
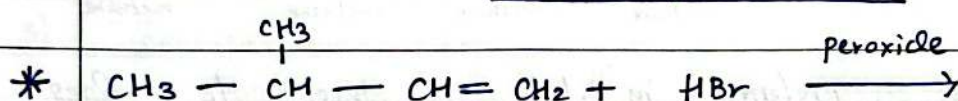


→ Carbocation stability order : $3^\circ > 2^\circ > 1^\circ$

governs which products forms.

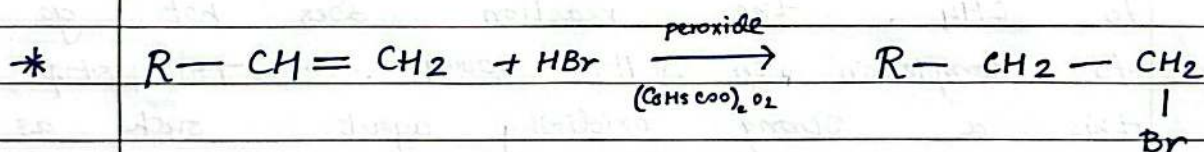
iii] Anti Markownikoff's Rule : Peroxide (Kharasch effect)

→ How it does not work



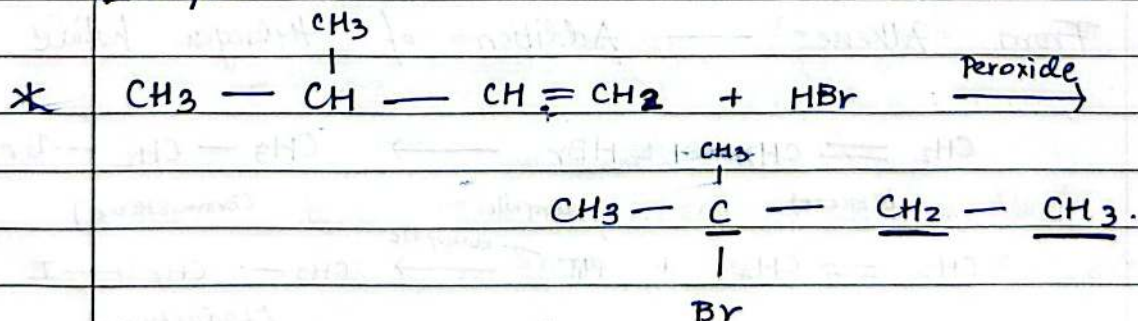
Wrong X.

→ How it does work

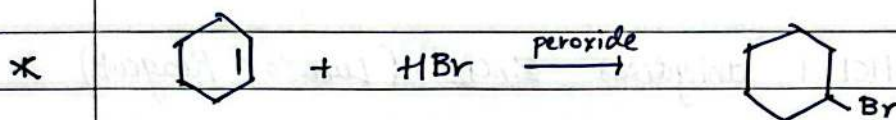
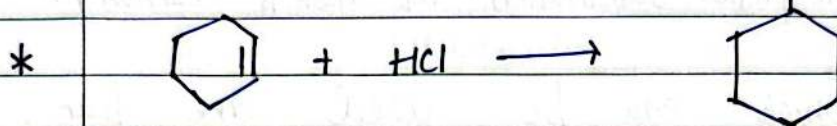
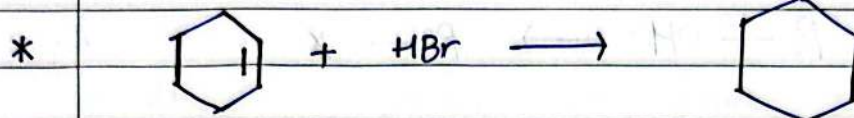


It only works for HBr (not HCl or HI).

Example: (1)

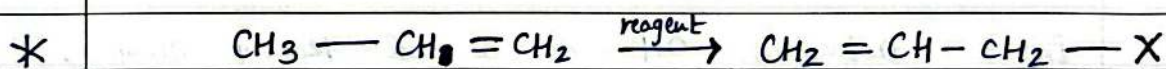


Example: (2)



iv) Allylic Substitution.

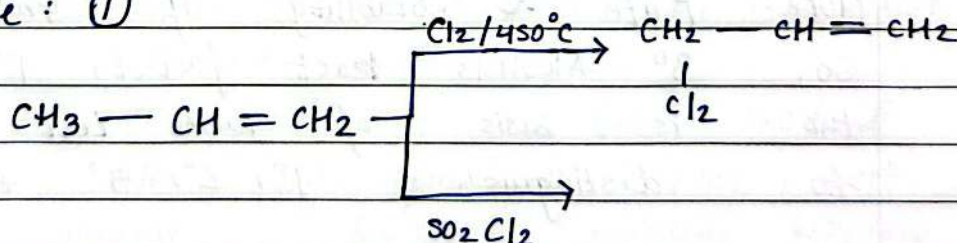
Unsaturated compounds can undergo substitution at the allylic position (the carbon to side of double bond) actually the double bond shifts



Reagents required for Allylic Substitution

- i) Halogen at high temperature ($400 - 500^\circ\text{C}$)
- ii) NBS (N-Bromosuccinimide) - allylic substitution by Br.
- iii) SO_2Cl_2 (sulfuryl Chloride) - allylic substitution by Cl.

Example: (1)



v1 From Alcohols

*



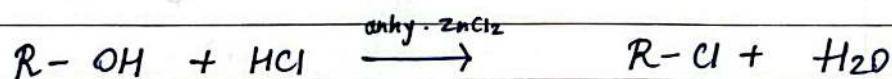
(Nucleophilic substitution at alcohols carbon)

Reagents which can be used are four.

i)

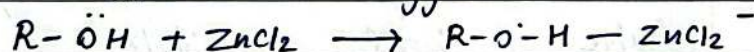
Conc. HCl + anhydrous $ZnCl_2$ (Lucas's Reagent)

*



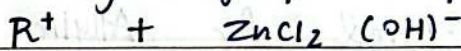
Mechanism

1) $ZnCl_2$ (a d block Lewis acid) accepts a lone pair from Alcohols oxygen:



this activates OH as a good leaving group.

2). Activated leaving group departs and forming:



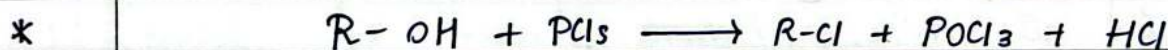
3). Cl then attacks the carbocation: $R^+ + Cl \longrightarrow R-Cl$

Note: Rate $\propto$ stability of Carbocation

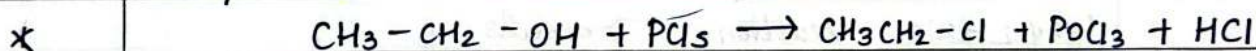
So, 3° Alcohols react fastest, 1° Alcohols slowest
this is basis of Lucas test, used to distinguish 1° , 2° , 3° Alcohols.

Alcohol	Example	Result
3° Alcohol	$(\text{CH}_3)_3\text{C}-\text{OH}$	white ppt of R-Cl — immediately
2° Alcohol	$(\text{CH}_3)_2\text{CH}-\text{OH}$	white ppt after 5-10 mins
1° Alcohol	$\text{CH}_3\text{CH}_2-\text{OH}$	No white ppt at room temp

ii) From PCl_5

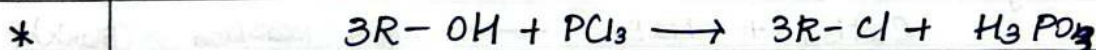


Example : ①

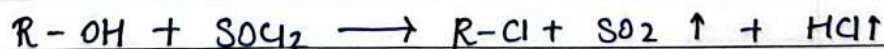


(PCl_5 is a solid that sublimes to gas on heating)

iii) From PCl_3



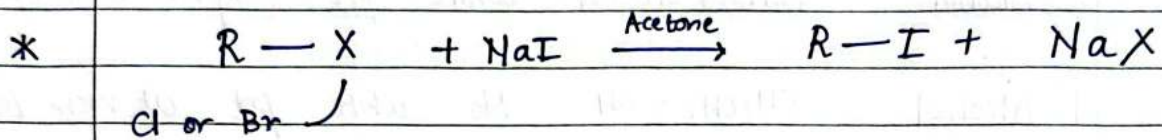
iv) From SOCl_2 (thionyl chloride) — Darzens process



This is considered the best method because all by products (SO_2 , HCl) are gaseous leaving reaction mixture leaving a clean purified R-Cl.

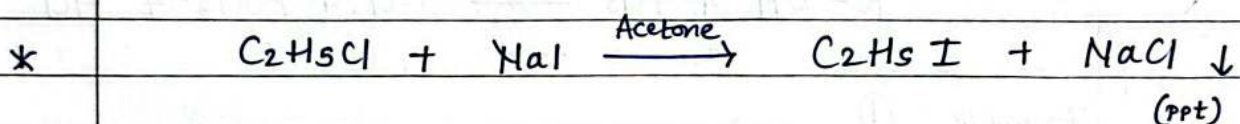
vi) Halogen Exchange Method.

1. Finkelstein Reaction — (exclusive for alkyl iodides)



It reacts very easily because of size of NaI

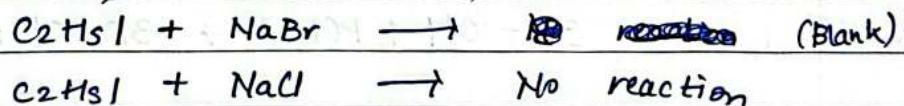
Example: ① $\text{NaI} > \text{NaBr} > \text{NaCl}$



Why this reaction proceeds

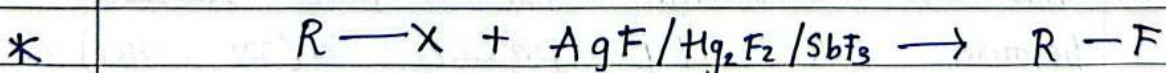
- NaI is more soluble in Acetone because of its big cation $\rightarrow$ ^{more} covalent character.
- NaI is more ionic and less soluble. It precipitates out as NaCl.

Why it does not work with NaBr or NaCl

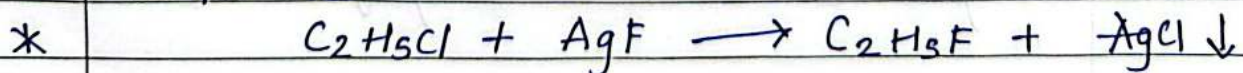


As Acetone is covalent, NaBr^- ⁰, NaCl^-

2. Swarts Reaction — (exclusively for alkyl fluorides)

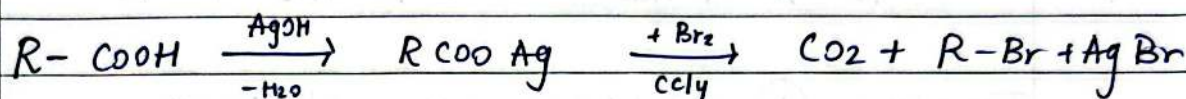


Example: ①



vii) Hunsdiecker Reaction — Borodine Reaction (exclusive for alkyl bromides)

→ silver salt of fatty acid (carboxylic acid)



i) The carboxylic acid is first converted to its silver salt.

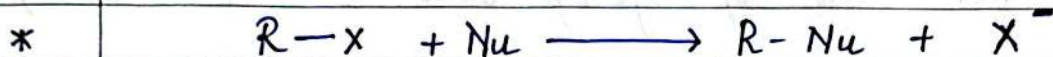
ii) which is then treated with Br_2 in CCl_4 the silver carboxylate loses CO_2 .

⇒ Physical Properties of Haloalkanes

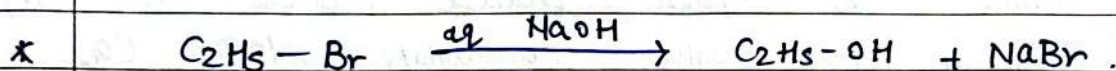
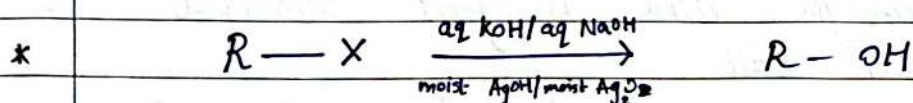
- State: lower alkyl halides (up to about 2 carbons for fluorides / chlorides) are gases most simple alkyl chlorides / bromides / iodides are colourless liquids, higher members are solids.
- Melting and boiling point: Increase with molecular mass and for a given alkyl group, increase down the halogen group $Cl < Br < I$ (heavier more polarisable halogen) Branching lowers b.p same trend as in alkanes
- Solubility: alkyl halides are practically insoluble in water but are soluble in organic solvents.
- Density: generally higher than corresponding alkane.

⇒ Chemical properties of Haloalkanes.

i) Nucleophilic Substitution:

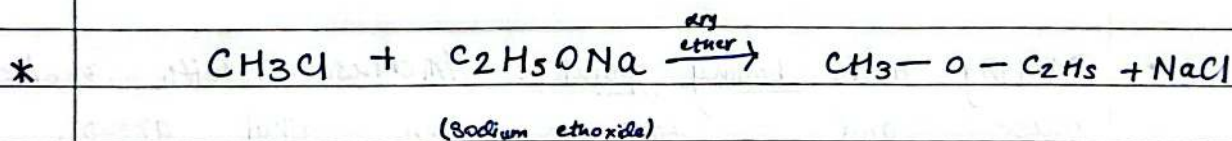
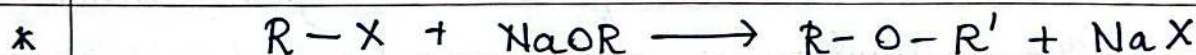


Example: ① Substitution by OH group
→ Formation of Alcohol

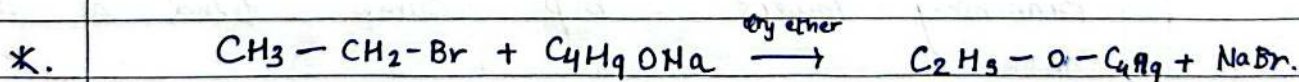


S_N2 mechanism, no carbocation formed
eg: $C_2H_5Br + NaOH \longrightarrow C_2H_5OH + NaBr$

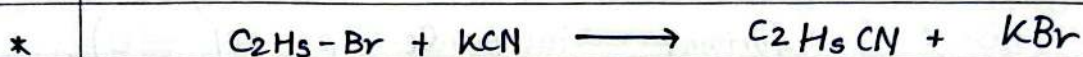
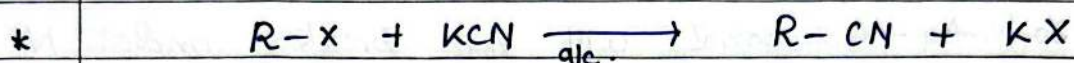
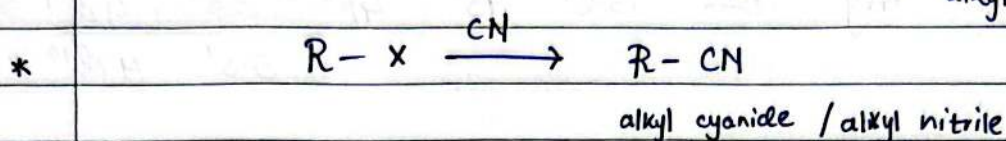
Example: ② Substitution by OR group
→ Formation of ether
[Williamson's Synthesis]



→ Williamson's Synthesis [Alkyl halide must be 1°]

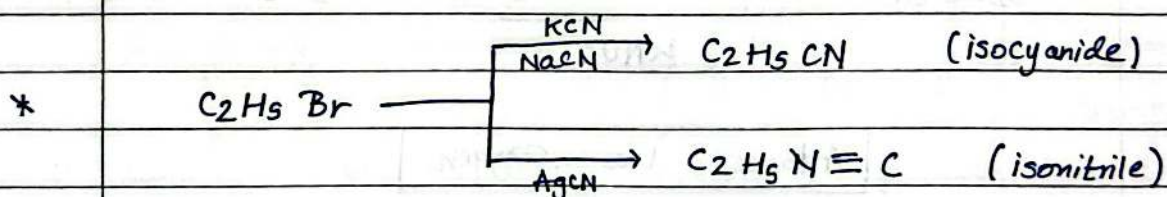
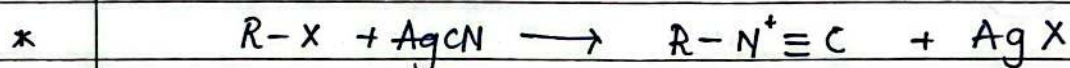
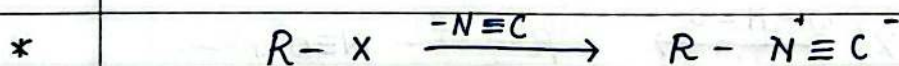


~~Imp~~ Example: (3) Substitution by -CN group.
 → formation of alkyl cyanide / alkyl nitrile



Attack occurs via carbon of CN; called ascent of carbon since the product has one more carbon than R-X

Example: (4) Substitution by -NC group.
 → formation of alkyl isocyanide / alkane isonitrile

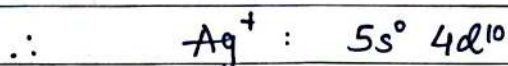
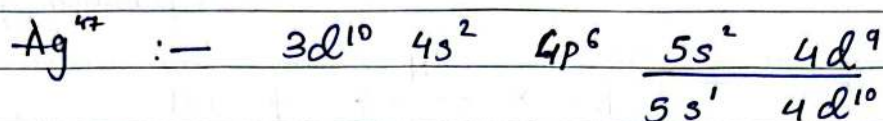


K-CN: Ionic bond	Ag-C≡N Covalent bond
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~~Imp~~ Fajan's Rule :
 • → K-CN is an ionic bond; the cyanide ion attacks through carbon because carbon carries a high electron density in free CN⁻ ion

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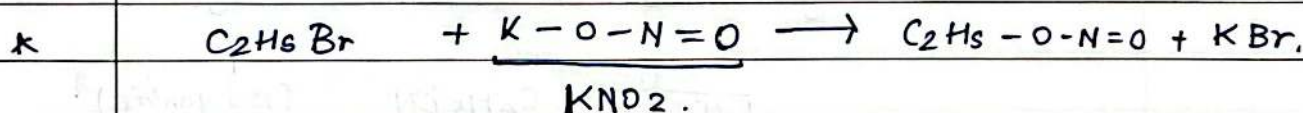
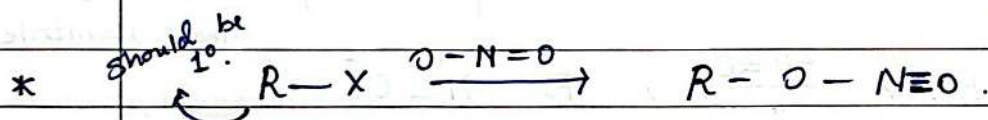
• $\rightarrow$ Ag-CN is a covalent bond



So, Ag-C bond will not break, and N will get attached to it. attacking the R-X forming isocyanide ($\text{R}-\text{N} \equiv \text{C}$).

• $\rightarrow$ Fajan's Rule - pseudo-noble gas config of cation increases covalent character / bond strength.

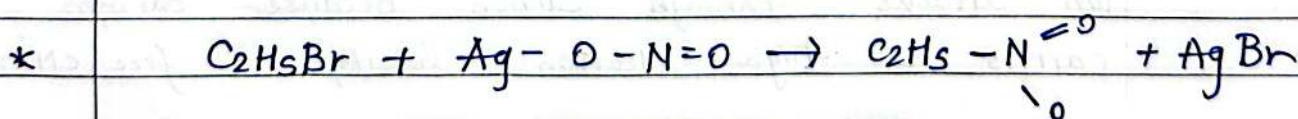
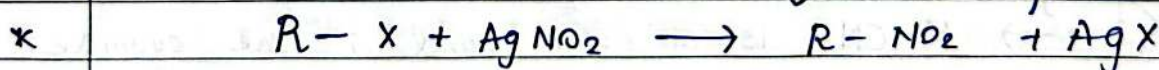
Example: (5) Substitution by $-\text{O}-\text{N}=\text{O}$ (nitrite)
 $\rightarrow$ formation of alkyl nitrite (ester)



Attacks via oxygen

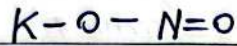
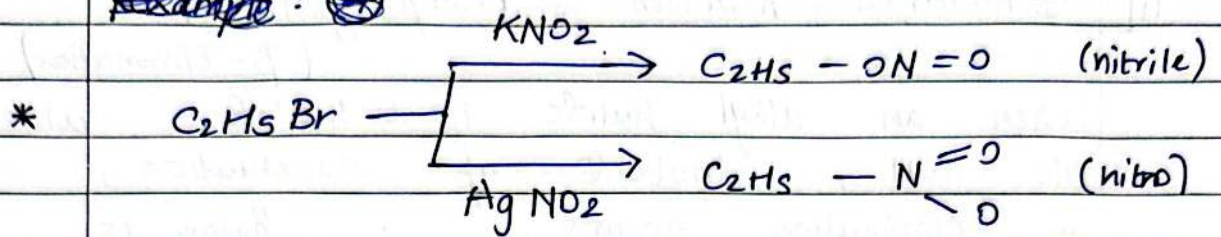
Example: (6) Substitution by $-\text{N} \begin{smallmatrix} \text{=O} \\ \text{O}^- \end{smallmatrix}$ (nitro gp)

$\rightarrow$ formation of nitro alkane



Attacks via nitrogen

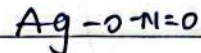
~~Example~~ : (7)



Ionic bond

(no pseudo

noble gas configuration / makes nitrile)

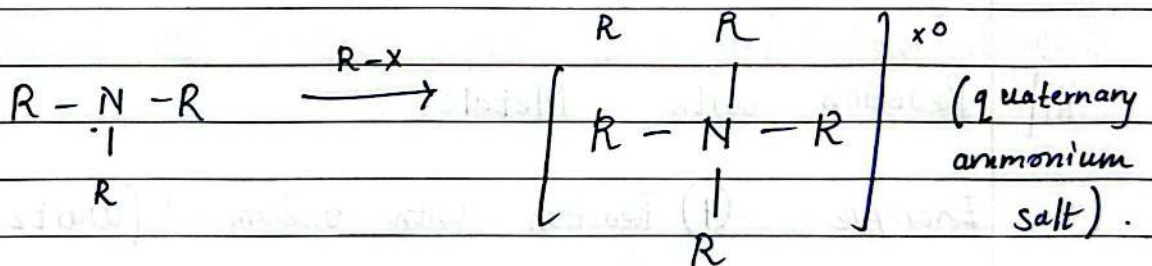
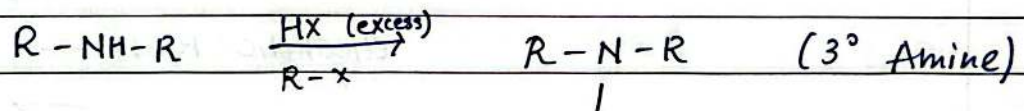
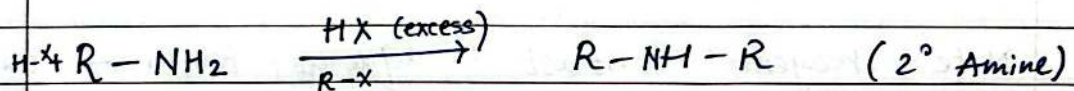
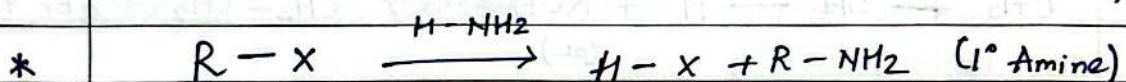


covalent bond.

(Follows Fajans Rule

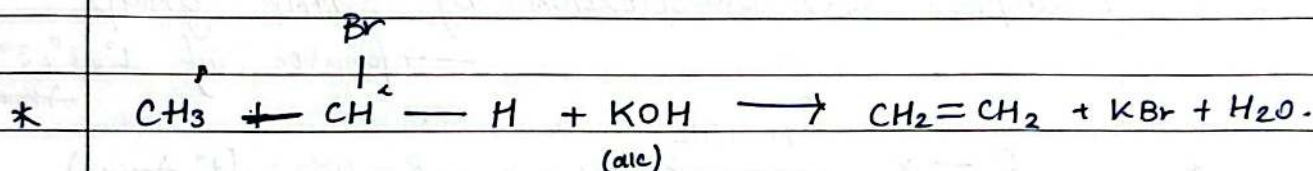
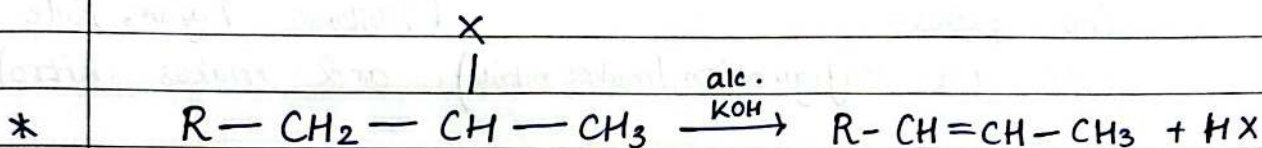
and makes nitro).

Example : (7) Substitution by $-NH_2$ group
 $\rightarrow$ formation of $1^\circ, 2^\circ, 3^\circ$ Amine



ii) Elimination Reaction (Saytzeff Rule) (β -Elimination)

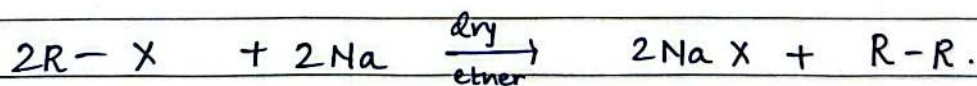
When an alkyl halide is heated with alc KOH, instead of substitution, an elimination occurs, a hydrogen is removed from β -carbon forming an Alkene.



Note reagent contrast : aqueous KOH $\rightarrow$ substitution
 $\rightarrow$ alcohol
 alcoholic KOH $\rightarrow$ elimination
 $\rightarrow$ alkene.

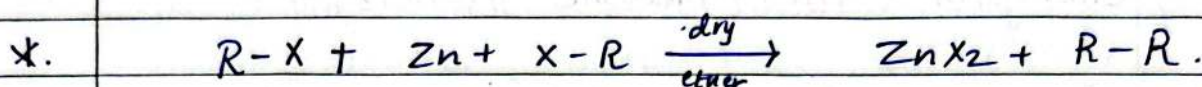
iii) Reaction with Metals.

Example: (1) Reaction with sodium (Wurtz Reaction).

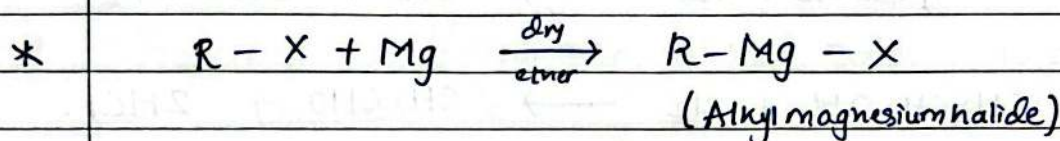


Couples two alkyl groups works best for identical R groups.

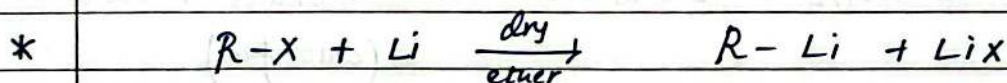
Example : (2) Reaction with Zinc
(Frankland Reaction).



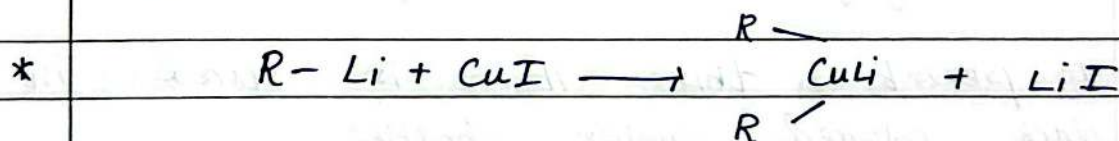
Example : (3) Reaction with Magnesium
(Grignards Reagent).



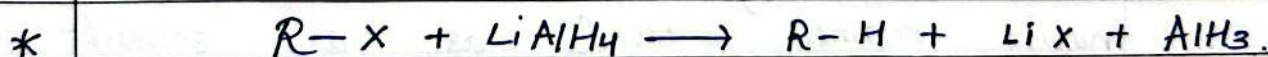
Example : (4) Reaction with Lithium
(Organolithium formation).



Example : (5) Reaction with $[Li+CuI]$ Gilman Reagent.



iv) Reaction with $LiAlH_4$

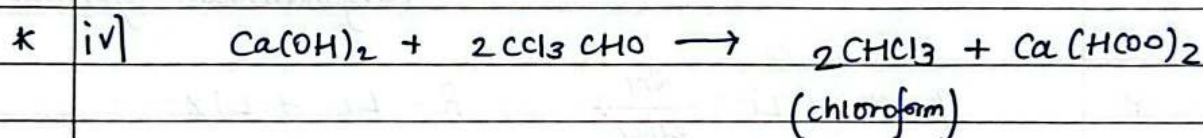
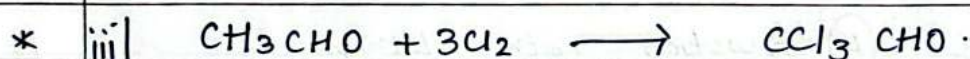
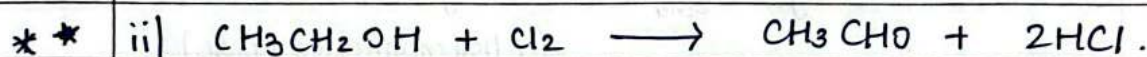
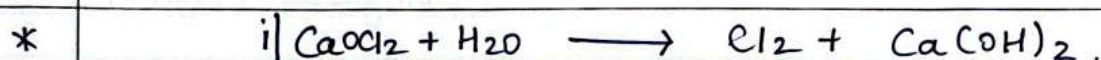


$LiAlH_4$ acts as a source of hydride ion which displaces the halide in an SN_2 substitution. reducing alkyl halide to parent alkane.

⇒ Chloroform, Iodoform, Freons and DDT

⇒ Chloroform (trichloromethane, CHCl_3)

- Preparation: by the haloform reaction.
ethanol or acetone is treated with bleaching powder (CaOCl_2).



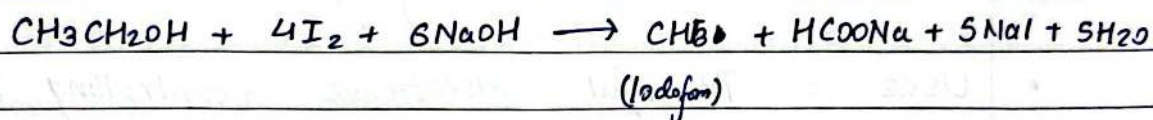
- Properties: Colourless, Sweet smelling liquid.
Slowly oxidised by air and light
to highly poisonous gas.

To prevent this it is stored in dark coloured amber bottles

- Uses: Historically used as an anaesthetic
now mainly used as a solvent and
in the manufacture of freons.

⇒ Iodoforms (triiodomethane, CHI_3).

- Preparation : by haloform reaction
ethanol treated with I_2 and NaOH .



- Properties : Yellow crystalline solid with a characteristic Antiseptic Smell.
- Use : was once used as an Antiseptic for wound dressing (the smell is called ^{hospital} smell)

⇒ Freons

- Structure : Chlorofluorocarbons (CFC's) - general formula $\text{C}_x\text{Cl}_y\text{F}_z$ eg: CFCl_3 (Freon-11), CF_2Cl_2 (Freon 12)
- Uses : refrigerants and aerosol propellants.
- Environmental effect : highly stable and unreactive in the lower atmosphere. and in upper reacts and depletes ozone layer.

⇒ DDT (Dichloro diphenyl trichloro ethane)

- Structure: two chlorobenzene rings attached to a central carbon that also bears a $-CCl_3$ group - i.e. $(4-ClC_6H_4)_2-CH-CCl_3$
- Uses: Powerful insecticide, controlling insect borne diseases.

⇒ Haloarenes.

⇒ Basic Idea and nature of C-X bond.

Haloarenes (aryl halides) have the halogen directly attached to a benzene ring carbon. examples - (chlorobenzene) C_6H_5Cl



Halogen atoms attached to the ring have a lone pair that can delocalise into the ring by resonance.

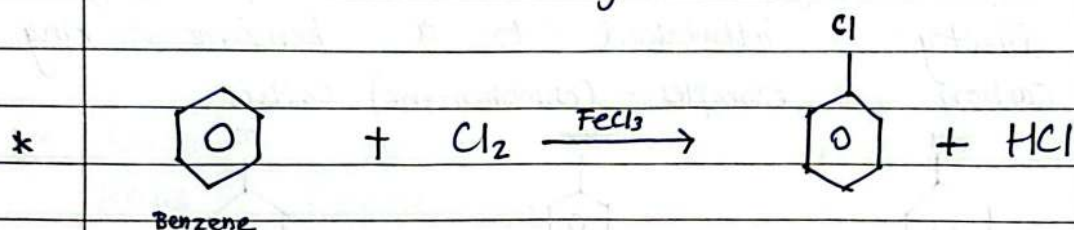
For halogens, -I effect dominates over +M effect.

⇒ Why the C-X bonds in haloarenes is stronger / shorter than haloalkanes and resists nucleophilic substitution under normal conditions.

1. The ring carbon is sp^2 hybridised (vs sp^3 in alkyl halides). higher s character means the bonding orbital holds electron density closer to nucleus, giving a shorter stronger C-X bond (electronegativity - $sp > sp^2 > sp^3$)
2. Resonance delocalises the halogens lone pair into the ring giving C-X bond partial double bond character.
3. The ring itself sterically shields carbon from backside nucleophilic attack.

⇒ Preparation of Haloarenes.

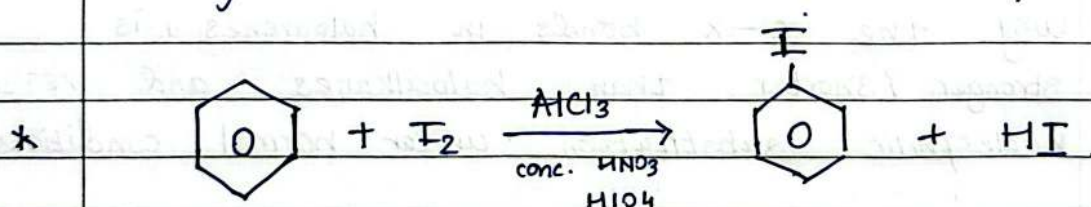
ii) From Hydrocarbons - Electrophilic Substitution
direct halogenation.



Any of several Lewis acid catalysts can be used:
FeCl₃, FeBr₃, FeCl₂, FeBr₂, AlCl₃, AlBr₃, ZnCl₂, BF₃, ZnBr₂.

Iodination

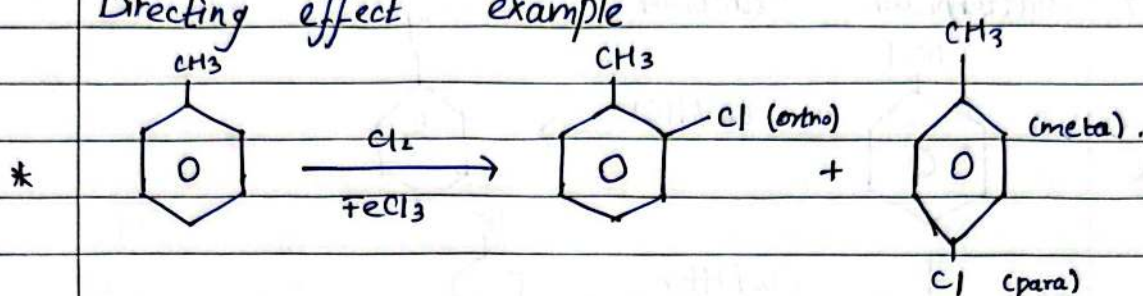
It requires an oxidizing agent since the reaction is reversible and HI a strong reducing which would reduce the product back



Note: Fluorobenzene can not be prepared by this method. as the bond formed between F and ring is far too reactive.

It is prepared by Balz-Schiemann Reaction.

Directing effect example

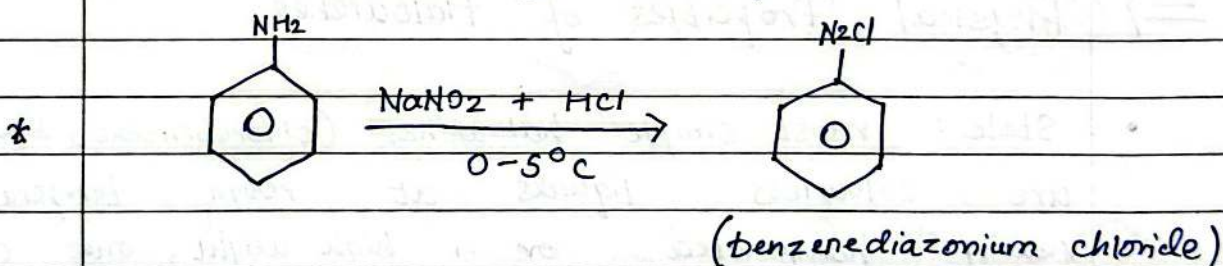


→ Directive influence of groups: A group already present on the benzene ring decides the position to be occupied by an incoming group on the ring.
Here: CH₃ = Directing group (ortho and para).

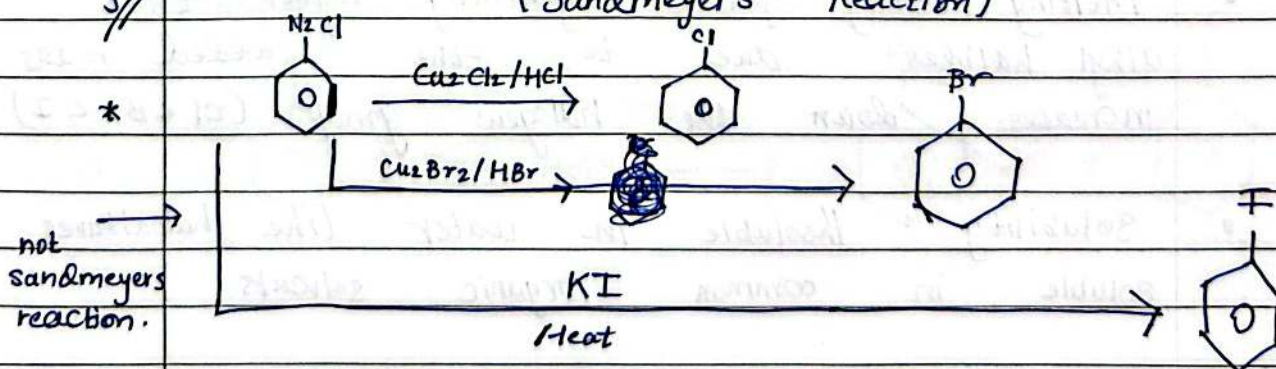
ii) From Diazonium Salts.

~~Imp~~ Step 1: Diazotization (from aniline).

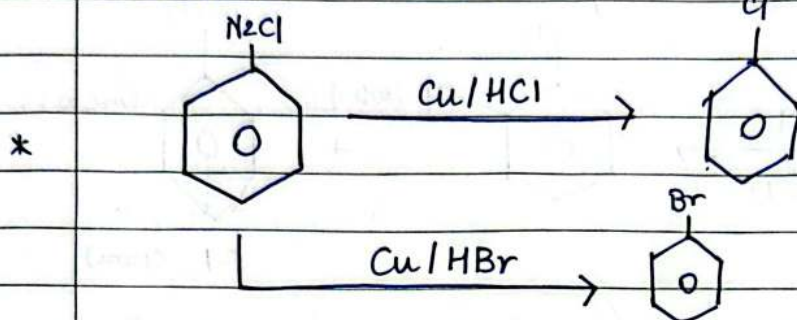
When a primary aromatic amine reacts with NaNO₂ + HCl at 0-5°C it forms the diazonium salt of Benzene.



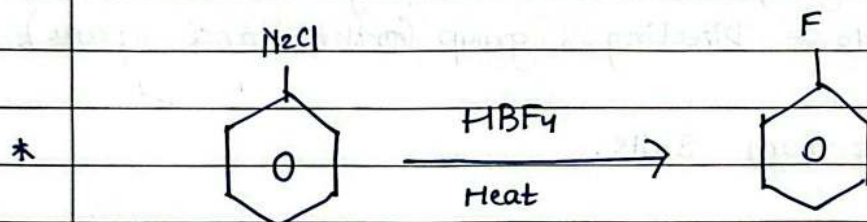
~~Imp~~ Step 2: Convert the diazonium salt to aryl halide (Sandmeyer's Reaction)



→ Quattermann Reaction



→ Balz - Schiemann Reaction.



the standard route to fluorobenzene since direct fluorination doesn't work

⇒ Physical Properties of Haloarenes.

- State: most simple haloarenes (chlorobenzene, brom...) are colourless liquids at room temperature heavily halogenated or high weight ones can be solids.
- Melting / boiling points: generally higher than alkyl halides due to the added mass increases down the halogen group ($\text{Cl} < \text{Br} < \text{I}$)
- Solubility: Insoluble in water like haloalkanes soluble in common organic solvents.

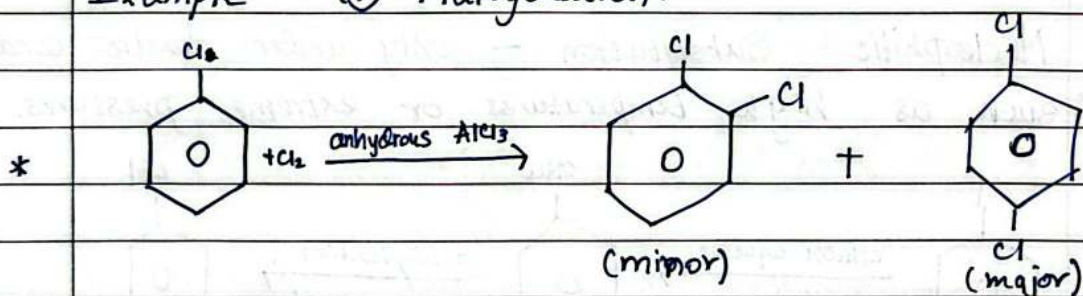
⇒ Chemical Properties of Haloarenes.

i) Electrophilic Substitution Reactions:

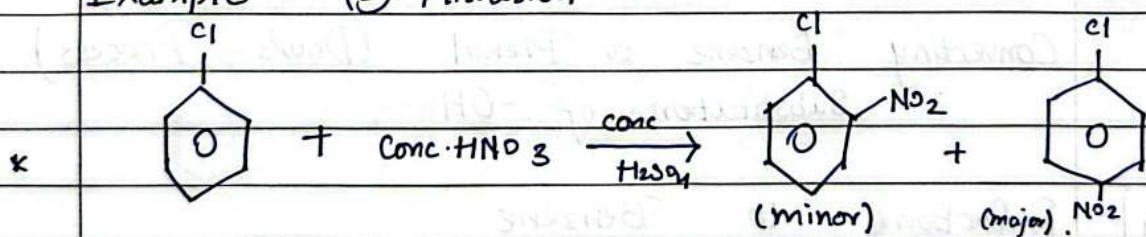
Directive influence of the halogen: although halogens are net deactivating ($-I$ dominates, $+M$) they are still ortho/para directing the $+M$ effect while smaller than $-I$ still selectively stabilises the arenium ion intermediate best.

Halogens are deactivating groups for electrophilic substitution, but if somehow the electrophile is forced to react it will at ortho and para.

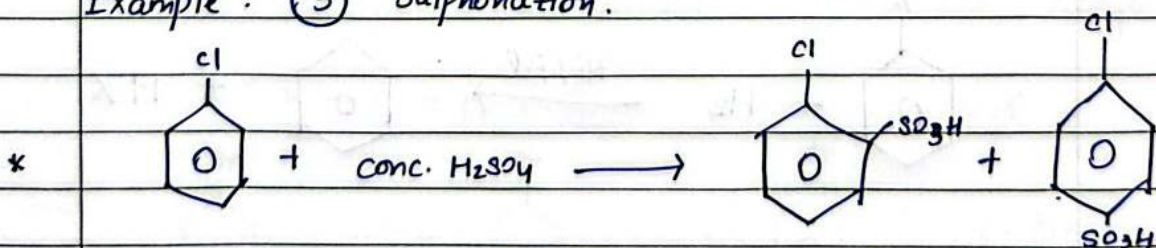
Example: ① Halogenation.



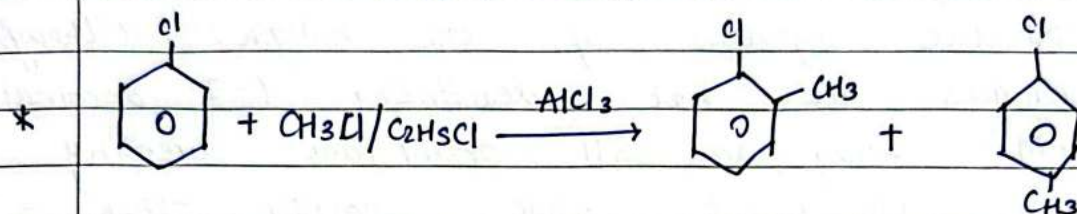
Example: ② Nitration



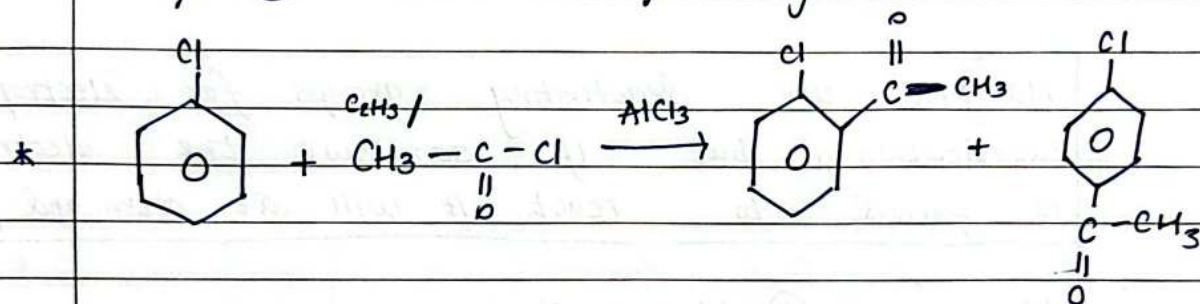
Example: ③ Sulphonation.



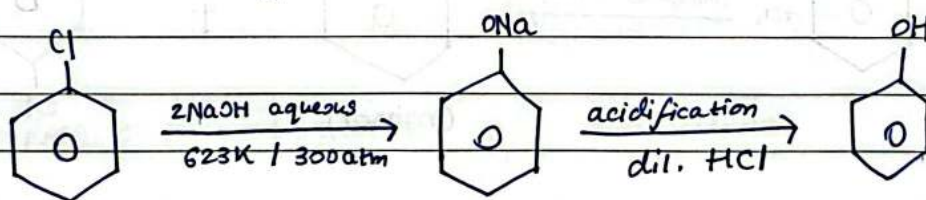
Example: (4) Friedel - Crafts Alkylation
(CH_3 , C_2H_5 , R^+)



Example: (5) Friedel crafts Acylation.

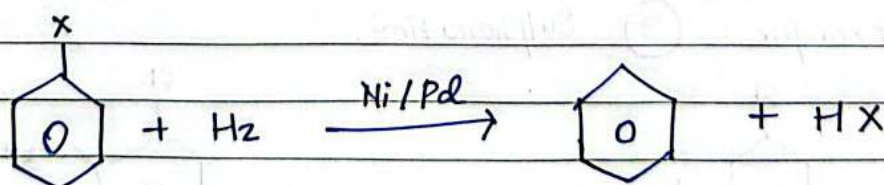


ii) ~~Imp~~ Nucleophilic Substitution - only under drastic conditions such as high temperatures or extreme pressures.



Converting Benzene to Phenol (Dow's Process)
Substitution of $-\text{OH}$

iii) Reduction to Benzene

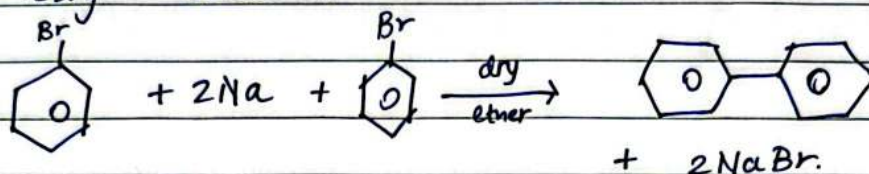


iv) Reaction with Metals

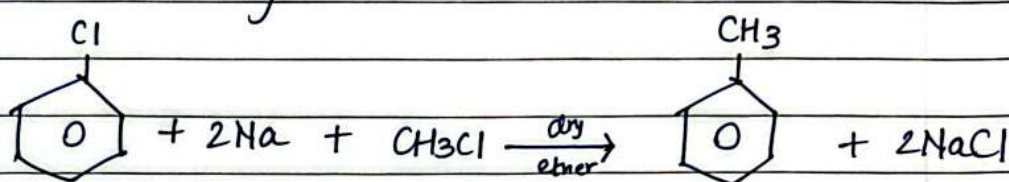
Imp

Example : (1) with sodium

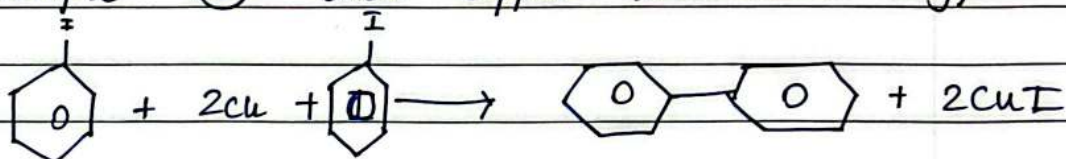
- Fittig Reaction



- Wurtz Fittig Reaction

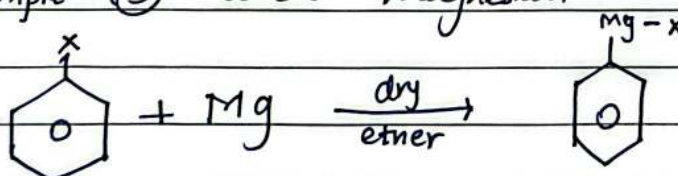


Example (2) with Copper (Iodobenzene only)



- Ullmann Reaction - this works for iodobenzene only

Example : (3) with magnesium (Grignard's Reagent)



Example : (4) with lithium (Gilman Reagent)

